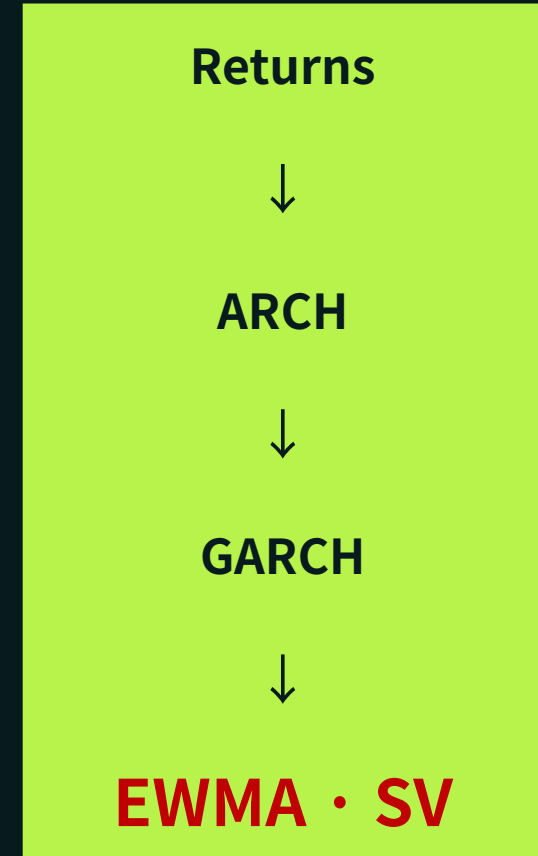


Hand-Calculation Workbook

ARCH and GARCH volatility in nine worked problems



Nine problems retrace the main deck's calculations in order

Units — returns in %, variance in %², annualization factor $\sqrt{252} \approx 15.87$

W1	————	Returns, annualizing	Getting units right Log return · $\sqrt{252}$
W2	————	ARCH and GARCH updates	Today's variance W2 ARCH(1) · W3 GARCH(1,1)
W4	————	Long run, k-day forecast	Mean reversion W4 V_L , half-life · W5 k-day
W6	————	EWMA vs. GARCH, 30-day vol	Main-deck example W6 3.12% vs 2.62% · W7 30-day
W8	————	SV step, LM test	Uncertainty and testing W8 SV range · W9 χ^2 test

Read each problem as: given → solution → answer

W1. 1% daily volatility is 15.9% annual volatility

A price rises from 100 to 102. With 1% daily volatility, what is annual volatility?

Price yesterday

100

Price today

102

Daily vol

1%

$\sqrt{252}$

≈ 15.87

Solution

$$r = \ln \frac{102}{100} = \ln 1.02 \approx 0.0198 \text{ (1.98\%)}$$

$$\sigma_{\text{annual}} = 1\% \times \sqrt{252} \approx 1\% \times 15.87 = 15.9\%$$

Answer: return $\approx 1.98\%$, annual vol $\approx 15.9\%$ — variance scales by 252, std. dev. by $\sqrt{252}$

W2. In ARCH(1), a big shock lifts today's variance over tenfold

$\alpha_0 = 0.2$, $\alpha_1 = 0.5$. Today's volatility after a -3% day versus a $+0.5\%$ day?

α_0
0.2

α_1
0.5

Shock day r
 -3%

Calm day r
 $+0.5\%$

Solution — (A) shock day · (B) calm day · (C) long-run

$$(A) \quad \sigma_t^2 = 0.2 + 0.5 \times (-3)^2 = 0.2 + 4.5 = 4.7, \quad \sigma_t = \sqrt{4.7} \approx 2.17\%$$

$$(B) \quad \sigma_t^2 = 0.2 + 0.5 \times (0.5)^2 = 0.2 + 0.125 = 0.325, \quad \sigma_t \approx 0.57\%$$

$$(C) \quad \bar{\sigma}^2 = \frac{0.2}{1 - 0.5} = 0.4, \quad \bar{\sigma} \approx 0.63\%$$

Answer: 2.17% vs. 0.57% — only the size (square) matters, not the sign

W3. A GARCH(1,1) update is a single sum of three terms

$\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$. Yesterday's variance 1.0 (%²), shock -3%. Today's vol?

 ω **0.02** α **0.10** β **0.88** σ^2 / ε yesterday**1.0 / -3**

Solution

$$\begin{aligned}\sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 = 0.02 + 0.10 \times 9 + 0.88 \times 1.0 \\ &= 0.02 + 0.90 + 0.88 = 1.80\end{aligned}$$

$$\sigma_t = \sqrt{1.80} \approx 1.34\% \Rightarrow 1.34\% \times 15.87 \approx 21.3\%$$

Answer: 1.34% daily, $\approx$ 21.3% annual — news (0.90) and memory (0.88) weigh about equally

W4. Same parameters: 15.9% long-run vol, half-life of about 34 days

Using W3's parameters, find the long-run variance V_L and the shock half-life

 ω **0.02** $\alpha + \beta$ **0.98** $1 - \alpha - \beta$ **0.02** $\ln 0.5$ **-0.693**

Solution

$$V_L = \frac{0.02}{1 - 0.10 - 0.88} = \frac{0.02}{0.02} = 1.0 \Rightarrow 1\% \text{ daily} \approx 15.9\% \text{ annual}$$

$$\text{half-life} = \frac{\ln 0.5}{\ln 0.98} = \frac{-0.693}{-0.0202} \approx 34 \text{ days}$$

Answer: $V_L = 1.0$ (%²), half-life ≈ 34 days — today's 21.3% drifts back to 15.9%

W5. The excess variance shrinks by 0.98 each day ahead

Starting from W3's tomorrow variance of 1.80, forecast 11 and 35 days ahead

$$\sigma_{+1}^2$$

1.80

$$V_L$$

1.0

$$0.98^{10}$$

≈ 0.817

$$0.98^{34}$$

≈ 0.503

Solution

$$E_t[\sigma_{t+k}^2] = 1.0 + 0.98^{k-1} \times (1.80 - 1.0)$$

$$k = 11 : 1.0 + 0.817 \times 0.8 = 1.65 \Rightarrow \sigma \approx 1.29\%$$

$$k = 35 : 1.0 + 0.503 \times 0.8 = 1.40 \Rightarrow \sigma \approx 1.18\%$$

Answer: 1.29% → 1.18% → ... → 1.00% — after one half-life the excess is halved

W6. EWMA reacts more than GARCH to the same shock

Main-deck example — vol 2% (variance 0.0004), return +10% (square 0.01), long-run 0.0004

EWMA λ

0.94

GARCH $\gamma / \alpha / \beta$

.03 / .03 / .94

$\omega = \gamma \times V_L$

0.000012

u^2

0.01

EWMA

$$\begin{aligned}\sigma^2 &= 0.94 \times 0.0004 + 0.06 \times 0.01 \\ &= 0.000376 + 0.000600 = 0.000976 \\ \sigma &= \sqrt{0.000976} \approx 3.12\%\end{aligned}$$

GARCH(1,1)

$$\begin{aligned}\sigma^2 &= 0.000012 + 0.03 \times 0.01 + 0.94 \times 0.0004 \\ &= 0.000012 + 0.000300 + 0.000376 = 0.000688 \\ \sigma &= \sqrt{0.000688} \approx 2.62\%\end{aligned}$$

Ten days on: EWMA stays at 3.12%; GARCH falls to 2.49%, heading for its 2% long run

※ GARCH 10 days ahead: $0.0004 + 0.97^9 \times (0.000688 - 0.0004) \approx 0.000619 \rightarrow \sqrt{} \approx 2.49\%$

W7. 30-day average volatility averages thirty variance forecasts

Volatility matched to a 30-day option — apply the geometric-series formula to W5's values

$$\sigma^2_{t+1} - V_L$$

0.8

$$\alpha + \beta$$

0.98

$$0.98^{30}$$

≈ 0.545

$$n$$

30

Solution

$$\begin{aligned}\bar{\sigma}_{30}^2 &= V_L + (\sigma_{t+1}^2 - V_L) \frac{1 - (\alpha + \beta)^{30}}{30(1 - \alpha - \beta)} \\ &= 1.0 + 0.8 \times \frac{1 - 0.545}{30 \times 0.02} = 1.0 + 0.8 \times 0.758 = 1.61 \\ \bar{\sigma}_{30} &\approx 1.27\% \Rightarrow \times 15.87 \approx 20.1\%\end{aligned}$$

Answer: 30-day average vol $\approx 20.1\%$ annual — between tomorrow (21.3%) and the long run (15.9%)

W8. In SV, today's volatility remains a range even given yesterday

$\mu_h = 0$, $\phi = 0.95$, yesterday's $h = 0.4$, s.d. of $\eta = 0.2$ — today's vol $\sigma_t = e^{(h_t/2)}$

μ_h

0

ϕ

0.95

h yesterday

0.4

s.d. of η

0.2

Solution

$$E[h_t] = 0 + 0.95 \times (0.4 - 0) = 0.38 \Rightarrow \sigma_t = e^{0.38/2} = e^{0.19} \approx 1.21\%$$

$$h_t = 0.38 \pm 0.20 \Rightarrow \sigma_t \in [e^{0.09}, e^{0.29}] \approx [1.09\%, 1.34\%]$$

Answer: center 1.21%, ± 1 s.d. range 1.09–1.34% — GARCH gives a number, SV a distribution

W9. An LM statistic above the critical value signals ARCH effects

Regressing squared residuals on 5 lags gives $T = 1,000$ and $R^2 = 0.05$

T
1,000

R^2
0.05

Lags q
5

5% critical
11.07

Solution — H_0 : no ARCH effect

$$LM = T \cdot R^2 = 1000 \times 0.05 = 50$$
$$\chi^2_{5, 0.95} = 11.07 < 50 \Rightarrow \text{reject } H_0$$

Answer: $LM = 50 > 11.07$ — reject H_0 (no ARCH effect) and move on to GARCH

**Add variances,
take square roots for volatility,
and scale by $\sqrt{252}$ to annualize.**

**Back to the main deck
Re-read with the charts**

W6 and KOSPI figures come from the original course examples; other parameters are illustrative